

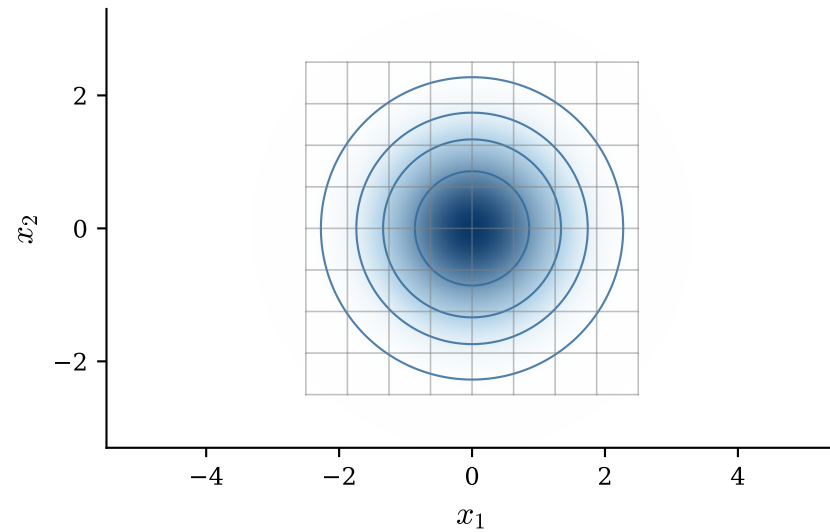
One density variation, two geometric costs

$$q = \mathcal{N}(0, I_2), \quad \phi_\theta(x_1, x_2) = (x_1 + \theta_1 \tanh x_1, \theta_2 x_2)$$

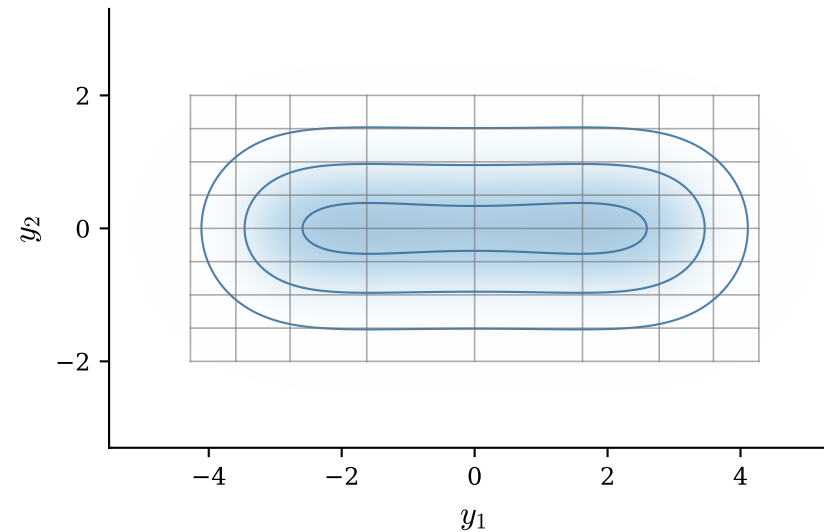
$$\theta = (1.8, 0.8), \quad d\theta = (0.9, 0.35), \quad v_{\theta, d\theta}(\phi_\theta(x)) = (d\theta_1 \tanh x_1, d\theta_2 x_2)$$

This induced velocity is a gradient: no WO projection is needed.

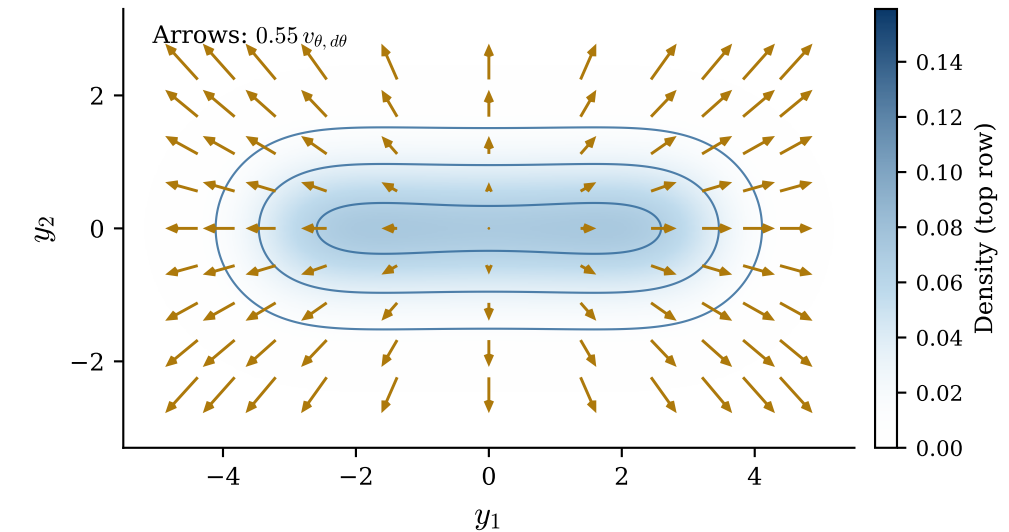
(a) Reference Gaussian and grid



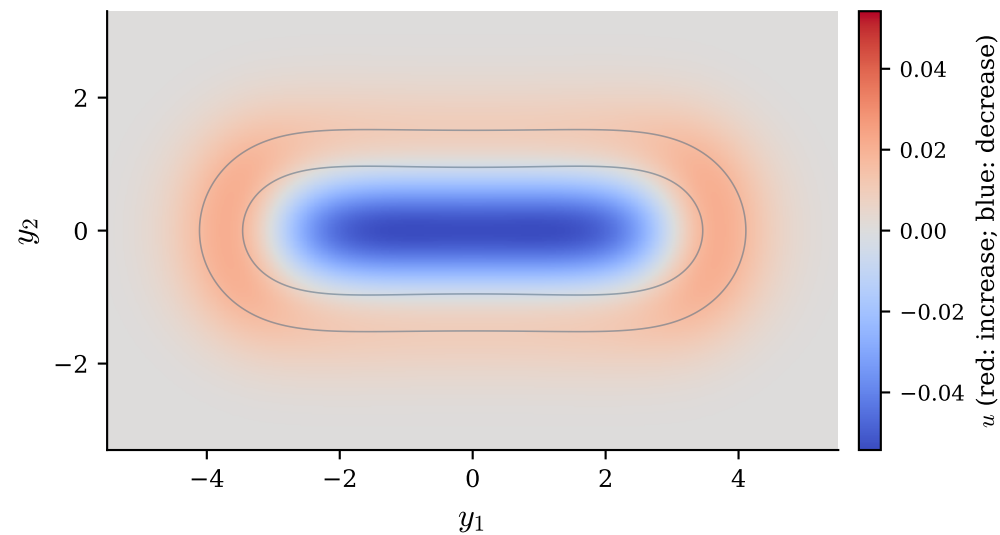
(b) Deformed density and grid



(c) Induced velocity on the density

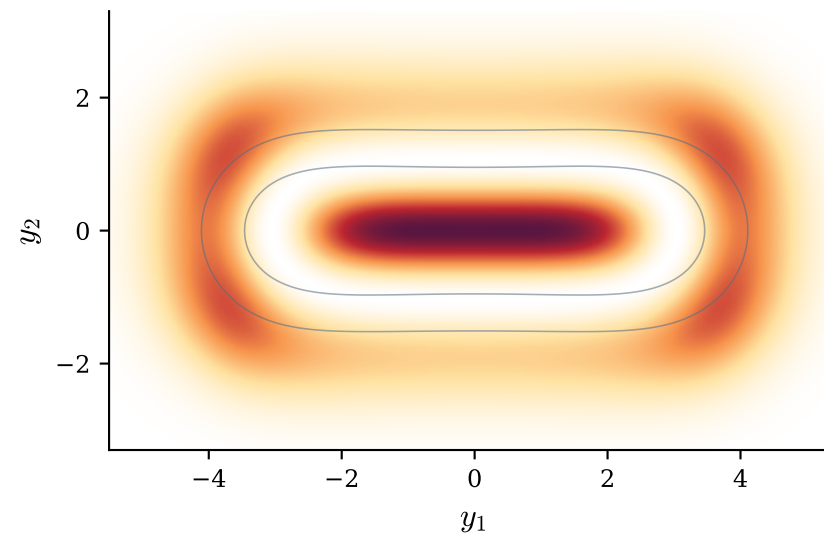


(d) Signed density variation



$$u = d_\theta p_\theta(d\theta) = -\text{div}_y(p_\theta v_{\theta, d\theta})$$

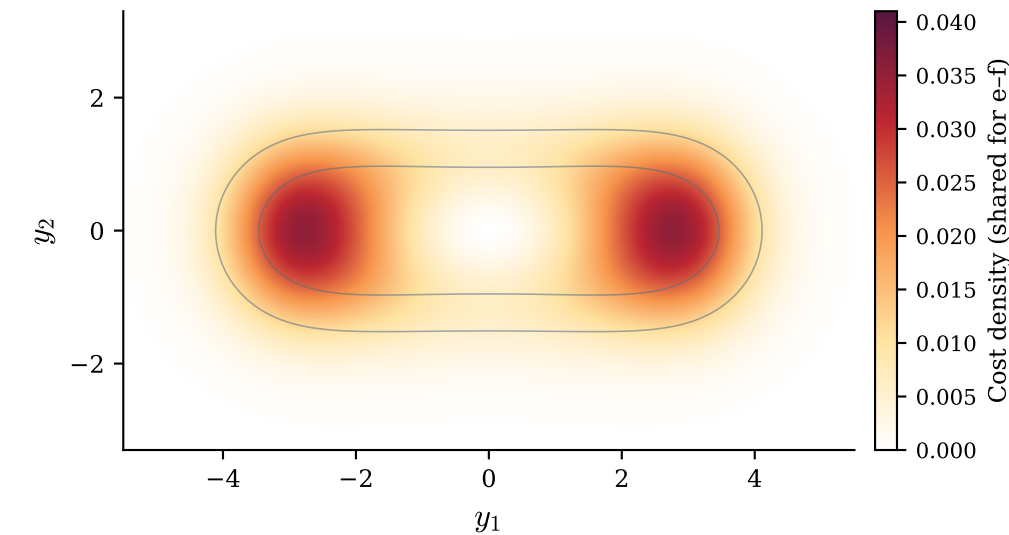
(e) Fisher-Rao cost density



$$\frac{[\text{div}_y(p_\theta v_{\theta, d\theta})]^2}{p_\theta}$$

Integral: 0.678

(f) Wasserstein-Otto cost density



$$p_\theta \|v_{\theta, d\theta}\|^2$$

Integral: 0.442

Fisher-Rao weights relative density change; WO weights the squared speed needed to move mass.

Identical spatial axes. The two cost maps share one color scale; their integrals are squared tangent norms, not distances.